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```
import numpy as np
import pandas as pd

df=pd.read_csv('Iris.csv')
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  -
0    sepal.length         150 non-null    float64
1    sepal.width          150 non-null    float64
2    petal.length         150 non-null    float64
3    petal.width          150 non-null    float64
4    variety               150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
df.variety.value_counts()
variety

Setosa      50
Versicolor  50
Virginica   50
Name: count, dtype: int64
df.head()
   sepal.length  sepal.width  petal.length  petal.width  variety
0           5.1           3.5           1.4           0.2   Setosa
1           4.9           3.0           1.4           0.2   Setosa
2           4.7           3.2           1.3           0.2   Setosa
3           4.6           3.1           1.5           0.2   Setosa
4           5.0           3.6           1.4           0.2   Setosa
features=df.iloc[:, :-1].values
label=df.iloc[:, 4].values

from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier

xtrain,xtest,ytrain,ytest=train_test_split(features,label,test_size=.2
,random_state=42)
model_KNN=KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain,ytrain)

KNeighborsClassifier()
```

```
print(model_KNN.score(xtrain,ytrain))
print(model_KNN.score(xtest,ytest))
0.9666666666666667
1.0

from sklearn.metrics import confusion_matrix
confusion_matrix(label,model_KNN.predict(features))

array([[50,  0,  0],
       [ 0, 47,  3],
       [ 0,  1, 49]])
```